

Lesson 40 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 items are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

- Q1. C.** NIC stands for Network Interface Card. A, B, and D are not expansions of the abbreviation.
- Q2. B.** LAN stands for Local Area Network. A, C, and D are not the correct full forms.
- Q3. B.** The client requests a service and the server provides it. A swaps the two roles; C describes neither role (forwarding without storing is not client-server); D swaps in the NIC and firewall roles.
- Q4. B.** One building or campus is a LAN. A PAN (A) covers one person, a MAN (C) covers one city, and a WAN (D) spans countries or the world.
- Q5. C.** Branches linked across one city form a MAN. A PAN (A) is one person's devices, a LAN (B) is one building, and D misdescribes both the medium and the range of a WLAN.
- Q6. C.** The worldwide public network is the Internet, not an intranet. Being private to one organisation (A), restricted to its staff (B), and carrying the internal portal (D) are all true of an intranet.
- Q7. B.** A star joins every node to one central device (hub or switch). A bus (A) shares one cable, a ring (C) joins neighbours in a circle, and a mesh (D) uses many direct links.
- Q8. A.** In a bus every node shares one main cable, so one break fails the whole network. A star (B) isolates a spoke failure, a mesh (C) reroutes over other paths, and a tree (D) fails branch-wise on the backbone, not on one shared cable.
- Q9. B.** In a ring each node joins exactly two neighbours in a closed circle. A describes a star, C describes a mesh, and D confuses the topology with a transmission medium.
- Q10. C.** A mesh is the most reliable and the costliest because of its many direct links. A bus (A) is cheapest, a star (B) uses one centre, and a ring (D) is a single circle.
- Q11. D.** Stars joined through a central backbone form a tree. A bus (A) is one shared cable, a ring (B) is a neighbour circle, and a mesh (C) is many direct links.
- Q12. A.** Twisted pair, coaxial, and fibre-optic are the three guided (wired) media. B, C, and D each list unguided (wireless) media: radio waves, microwaves, satellite links, infrared, or Bluetooth.
- Q13. C.** Fibre-optic cable carries light signals and is the fastest of the three cables. Twisted pair (A) is copper and slowest, coaxial (B) is shielded copper, and D combines two unrelated items.

- Q14. A.** A hub broadcasts every incoming signal to all ports. A switch (B) forwards only to the addressed port, a router (C) paths packets between networks, and a firewall (D) blocks unauthorised access.
- Q15. C.** A switch forwards only to the addressed port. A hub (A) broadcasts to all, a repeater (B) only regenerates the signal, and a modem (D) converts digital–analogue for the line.
- Q16. B.** A repeater regenerates a weak signal without choosing a path. A router (A) chooses paths, a gateway (C) translates protocols, and a firewall (D) guards access.
- Q17. D.** A router joins different networks and chooses the best path per packet. A hub (A) broadcasts inside one network, a switch (B) forwards inside one network, and a bridge (C) joins two LAN segments.
- Q18. D.** All three statements are correct: a bridge joins two LAN segments with filtering (I), a gateway translates between different protocols (II), and a firewall blocks unauthorised access (III). A omits II and III; B omits III; C omits I.
- Q19. B.** An access point lets Wi-Fi devices join the wired network. A swaps in the firewall's role for the modem, C swaps in the repeater's role for the router, and D swaps in the gateway's role for the NIC.
- Q20. B.** A modem (modulator-demodulator) converts between digital computer signals and analogue line signals. A switch (A) forwards inside the LAN, a bridge (C) joins LAN segments, and an access point (D) admits Wi-Fi devices.
- Q21. A.** LAN is Local Area Network. B, C, and D are not the recognised expansion.
- Q22. A.** Routers link networks. B, C, and D are peripherals, not inter-network devices.
- Q23. A.** WANs cover the widest areas. LAN (B) and PAN (C) are smaller; D is not a network class.
- Q24. A, B, C.** Hubs, switches, and routers move network data. A toaster (D) does not.
- Q25. A, B, C.** The LAN/WAN split and router role are correct. D is false: printers do not route packets.
- Q26. C.** The Internet spans large geographic areas and interconnected networks, so it is a WAN. LAN, MAN, and PAN cover smaller scopes.
- Q27. A.** A router forwards packets between networks. A hub repeats local signals; keyboard is input and plotter is output.
- Q28. A.** In a star topology each node connects to a central hub/switch. Ring and bus use different arrangements; mesh uses multiple inter-node paths.
- Q29. A.** A Network Interface Card (NIC) provides network connectivity. The others are output or recognition devices.
- Q30. A, B, C.** LAN, router, and switch roles are correctly described. A hub repeats signals and does not route using IP addresses.